UNIT - 1

INTRODUCTION & PROPERTIES OF MATERIALS

History - Nanomaterials - Classification - Crystals - Band structure - Properties

1. History of Nano Science

Nanoscience is the study of phenomena and manipulation of matter at the atomic, molecular and macromolecular scale, where properties differ markedly from bulk. Nanotechnology is the application of this knowledge to design, produce and use useful devices.

Important milestones

Year	Contribution
~4th century AD	Lycurgus Cup (Rome): glass with Au-Ag nanoparticles looks green in reflected light, red in transmitted light
1857	Michael Faraday prepared colloidal gold (ruby red) and explained its colour is due to tiny particles
1905	Einstein estimated molecular size from sugar solution (about 1 nm)
1931	Ernst Ruska and Max Knoll invented the Transmission Electron Microscope (TEM)
1959	Richard Feynman lecture "There is plenty of room at the bottom" - idea of manipulating atoms
1974	Norio Taniguchi (Tokyo) first used the word "nano-technology"
1981	Binnig and Rohrer invented the Scanning Tunnelling Microscope (STM), Nobel Prize 1986
1985	Kroto, Curl and Smalley discovered fullerene C60 (Nobel 1996)
1986	Eric Drexler book "Engines of Creation"; Binnig, Quate, Gerber invented AFM

1991	Sumio Iijima discovered carbon nanotubes (CNTs) at NEC, Japan
2004	Geim and Novoselov isolated graphene using scotch tape (Nobel 2010)

Remember:
Feynman (1959) is called the "father of nanotechnology" for the vision; Taniguchi coined the term; Drexler popularised it.

2. Definitions

Nanometer

One nanometer is one billionth of a metre.

$$1\ nm = 10^{-9}\ m = 10\ Å = 10^{-3}\ \mu m$$

- Hydrogen atom diameter is about 0.1 nm; a DNA double helix is about 2 nm wide
- Human hair is about 80,000 nm thick; red blood cell about 7,000 nm
- Typical atom spacing in a solid is 0.2 – 0.3 nm, so 1 nm spans only 3 – 4 atoms

Nanomaterials

Materials having at least one dimension in the range 1 – 100 nm are called nanomaterials. Their size is comparable to the de Broglie wavelength / exciton Bohr radius of electrons, so quantum effects appear.

Nanotechnology

Design, characterisation, production and application of structures, devices and systems by controlling shape and size at the nanometre scale (1 – 100 nm).

Why is the nano scale special? (two main reasons)

1. **Surface to volume ratio increases** – for a sphere $S/V = 3/r$. Smaller r means a large fraction of atoms lie on the surface, so reactivity, catalysis, melting point etc. change.
2. **Quantum confinement** – when size is comparable to the electron

wavelength, energy levels become discrete and the band gap increases. Optical, electrical and magnetic behaviour changes.

Particle size	Total atoms	% atoms on surface
10 nm	30,000	~ 20 %
3 nm	4,000	~ 50 %
1 nm	30	~ 99 %

3. Classification of Nanomaterials

Based on the number of dimensions that are NOT confined to the nano range (i.e. free electron movement), nanomaterials are classified as 0D, 1D, 2D and 3D.

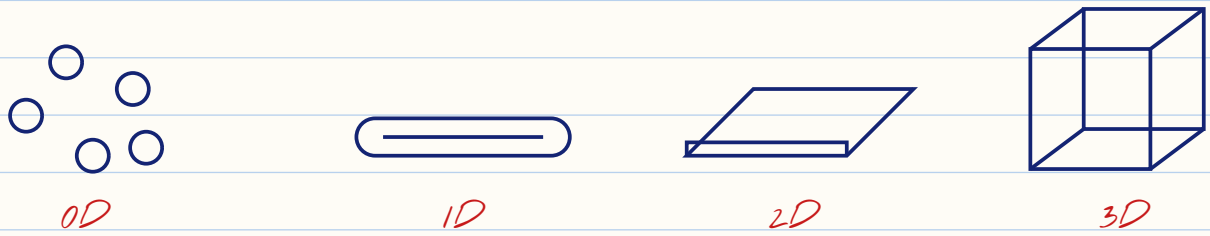


Fig: Dimensionality of nanomaterials (shape of the nano object)

Class	Confined in	Free in	Examples
0D	all 3 dimensions (x, y, z)	0 dimensions	Quantum dots, nanoparticles, fullerene C60, clusters
1D	2 dimensions	1 dimension (along length)	Nanowires, nanorods, nanotubes, nanofibres
2D	1 dimension (thickness)	2 dimensions	Graphene, thin films, quantum wells, nanosheets, MoS_2
3D	none (nano-structured bulk)	3 dimensions	Nanocrystalline bulk, nanocomposites, powders, multilayers, fullerite

Other ways of classifying

- **By material**: carbon-based (fullerene, CNT, graphene), metal-based (Au, Ag, Fe_3O_4), ceramic / metal oxide (TiO_2, ZnO), semiconductor (CdSe, CdS, Si), polymeric (dendrimers) and composite nanomaterials

- **By origin**: natural (volcanic dust, viruses, lotus leaf structure, gecko foot hairs), incidental (soot, welding fumes) and engineered (manufactured on purpose)
- **By structure (Gleiter)**: nanostructured materials are classified by shape (layered, rod, equiaxed) and chemical composition

4. Crystal Symmetries

A crystal = lattice + basis. A lattice is an infinite regular array of points; the basis is the atom or group of atoms attached to every lattice point. Unit cell: the smallest repeating block described by lattice parameters a, b, c and angles α, β, γ.

Symmetry operations

- **Translation** – shifting by a lattice vector leaves the crystal unchanged
- **Rotation** – only 1, 2, 3, 4 and 6-fold axes are allowed in crystals (5-fold cannot fill space without gaps)
- **Reflection (mirror plane)** – m
- **Inversion** – every point (x,y,z) goes to (−x,−y,−z), centre of symmetry
- **Rotation-inversion** and glide / screw operations (combined with translation) give the space groups

> **Fact:**
> 32 point groups and 230 space groups exist. Combining symmetry with lattice type gives 7 crystal systems and 14 Bravais lattices.

Seven crystal systems

System	Axes	Angles	Example
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$	NaCl, Cu, Si, Fe
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	TiO_2 (rutile), Sn
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	$BaSO_4$, Ga
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°, \gamma = 120°$	Zn, Mg, graphite, ZnO

Trigonal (Rhombohedral)	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	Bi, calcite, quartz
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90° \neq \beta$	Gypsum, monoclinic S
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	$CuSO_4.5H_2O$, $K_2Cr_2O_7$

14 Bravais lattices

- Cubic – 3: simple (SC), body-centred (BCC), face-centred (FCC)
- Tetragonal – 2: simple, body-centred
- Orthorhombic – 4: simple, base-centred, body-centred, face-centred
- Hexagonal – 1, Trigonal – 1
- Monoclinic – 2: simple, base-centred
- Triclinic – 1

Cubic cell	Atoms / cell	Coord. no.	Packing factor	Examples
SC	1	6	0.52	Polonium
BCC	2	8	0.68	Fe, Cr, W, Na
FCC	4	12	0.74	Cu, Al, Au, Ag
HCP	6	12	0.74	Mg, Zn, Ti

5. Crystal Directions and Planes

Crystal directions [u v w]

1. Choose the origin and draw the direction vector through it (or shift it to pass through the origin).
2. Take the coordinates of the point where the vector leaves the cell in units of a, b, c.
3. Multiply / divide by a common factor to get the smallest integers.
4. Enclose in square brackets [u v w]; negative index is written with a bar on top.

- x-axis = [100], y-axis = [010], z-axis = [001], body diagonal = [111], face diagonal = [110]

- Family of equivalent directions is written with angle brackets, e.g. ⟨100⟩ includes [100], [010], [001] and their negatives

Crystal planes – Miller indices (h K l)

Miller indices are three smallest integers that are the reciprocals of the intercepts of the plane on the three axes.

1. Find the intercepts of the plane on the x, y, z axes in terms of a, b, c (e.g. 2, 3, 1).
2. Take reciprocals (1/2, 1/3, 1).
3. Multiply by the LCM of the denominators (6): 3, 2, 6.
4. Enclose in round brackets: (3 2 6). A plane parallel to an axis has intercept ∞, so its index is 0.

> **Key:**
> Plane (hkl) cuts the axes at a/h, b/k, c/l. Family of planes is written {hkl}. In cubic crystals [hkl] is perpendicular to the plane (hkl).

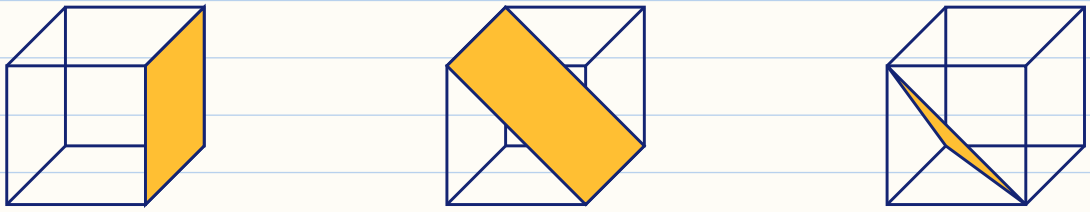


Fig: Important planes in a cubic lattice

Interplanar spacing

$$d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)} \qquad \text{(cubic)}$$

$$2 d \sin\theta = n \lambda \qquad \text{(Bragg's law)}$$

Example: for $a = 4$ Å, plane (111): $d = 4/\sqrt{3} = 2.31$ Å. For hexagonal: $1/d^2 = (4/3)(h^2 + hk + k^2)/a^2 + l^2/c^2$.

Hexagonal systems use four-index Miller-Bravais notation (h k i l) with $i = -(h + k)$.

Reciprocal lattice (needed for band structure)

- Reciprocal lattice vectors: $a^* = 2\pi (b \times c) / V$, etc. so that $a \cdot a^* = 2\pi$

- Every set of planes (hkl) in real space = one point in reciprocal space at distance $2\pi/d_{hkl}$
- **Brillouin zone** = Wigner-Seitz cell of the reciprocal lattice. First BZ of FCC is a truncated octahedron; of BCC is a rhombic dodecahedron. Symmetry points: Γ (centre), X, L, K.

6. Band Structure of Solids

Origin of bands

- An isolated atom has sharp energy levels. When N atoms are brought close together, wave functions overlap and, by Pauli exclusion, each level splits into N closely spaced levels
- These form a *quasi-continuous band*. Between bands lie forbidden gaps (E_g)
- Valence band (VB) = highest filled band. Conduction band (CB) = next empty / partly filled band

Bloch theorem

In a periodic potential $U(r + R) = U(r)$, solutions of the Schrodinger equation have the form:

$$\psi_K(r) = u_K(r)\, e^{i K \cdot r} \qquad \text{with } u(r + R) = u(r)$$

- K is the crystal wave vector; E(K) is periodic in K with period $2\pi/a$, so we only plot the first Brillouin zone ($-\pi/a$ to $+\pi/a$)
- **Kronig-Penney model** (rectangular periodic wells) gives: $\cos(Ka) = P \sin(\alpha a)/(\alpha a) + \cos(\alpha a)$. Allowed values are those where RHS lies between −1 and +1; others give forbidden gaps
- Near a band edge the effective mass $m^* = \hbar^2 / (d^2E/dK^2)$ replaces m: narrow band means heavy electrons

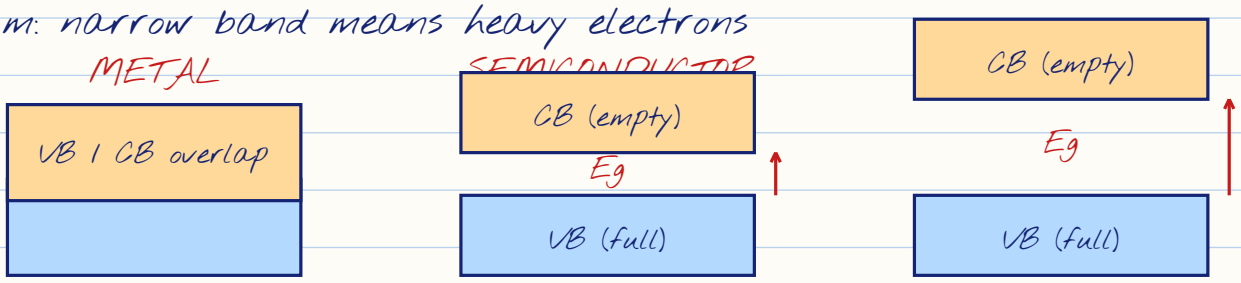


Fig: Band picture of metals, semiconductors and insulators

Property	Metal	Semiconductor	Insulator
Band gap Eg	0 (overlap / half filled)	0.1 – 3 eV (Si 1.12, Ge 0.67, GaAs 1.42)	> 3 eV (diamond 5.5, SiO2 ~9)
Resistivity (ohm m)	10^{-8}	10^{-5} to 10^{6}	10^{10} to 10^{20}
Temp. effect	Resistance rises with T	Resistance falls with T	Breakdown at high T

Direct and indirect band gap

- **Direct**: CB minimum and VB maximum at the same K (GaAs, InP, CdSe, ZnO). A photon alone can excite / recombine an electron so it is strongly emitting – used in LEDs and lasers
- **Indirect**: minimum and maximum at different K (Si, Ge). A phonon must help to conserve momentum, so emission is weak

$$E_F: \quad f(E) = 1 / [1 + \exp((E - E_F)/K_B T)]$$

Fermi level E_F lies inside the band for metals and near mid-gap for intrinsic semiconductors; n-type moves it towards the CB, p-type towards the VB.

7. Properties of Materials

For each property we study the bulk behaviour first and then how it changes when size drops to the nanometre scale.

7.1 Mechanical properties

- Defined by stress-strain behaviour: Young's modulus Y = stress/strain, hardness, yield strength, toughness, ductility
- Hardness and yield strength of nanocrystalline metals follow the Hall-Petch relation:

$$\sigma_y = \sigma_0 + K\, d^{-1/2}$$

- where d = grain size. Smaller grains mean more grain boundaries, which block dislocation motion, so the material is harder (Cu with 10 nm grains is about 5x harder than coarse Cu)
- Below d ≈ 10 – 20 nm the trend reverses (inverse Hall-Petch)

because grain-boundary sliding dominates

- Nano-ceramics become ductile / superplastic; CNTs have Young's modulus ≈ 1 TPa and tensile strength 50 – 100 GPa (about 100 times stronger than steel at 1/6 weight)
- Nanocomposites (clay-polymer, CNT-polymer) show large increase in strength with a very small filler fraction

7.2 Electrical properties

- Conductivity $\sigma = n e \mu = n e^2 \tau / m^*$. In bulk metals, resistivity comes from scattering by phonons, impurities and grain boundaries (Matthiessen's rule $\rho = \rho_T + \rho_0$)
- When the size becomes smaller than the electron **mean free path** (≈ 40 nm in Cu), surface and grain-boundary scattering raise the resistivity
- In very small conductors transport is ballistic; conductance is quantised in steps of $G_0 = 2e^2/h = 77.5\ \mu S$ (resistance 12.9 kΩ per channel)
- **Coulomb blockade**: charging energy $E_C = e^2/2C$ of a tiny island exceeds $k_B T$, so electrons tunnel one at a time (basis of the single-electron transistor)
- Carbon nanotubes can be metallic or semiconducting depending on chirality (n, m): metallic if (n – m) is a multiple of 3
- Semiconductor nanocrystals: band gap widens, carrier concentration changes, conductivity can be tuned by doping and size

7.3 Dielectric properties

- A dielectric is an insulator that polarises in an electric field. Polarisation $P = \varepsilon_0 \chi E$ and relative permittivity $\varepsilon_r = 1 + \chi$
- Mechanisms: electronic, ionic (atomic), orientational (dipolar) and space-charge (interfacial) polarisation. Each acts up to its own frequency limit
- Nano-structured dielectrics have a very large interface (grain-boundary) area, so **space-charge polarisation** rises and ε_r can become very high at low frequency (e.g. nano $BaTiO_3$)

$CaCu_3Ti_4O_{12}$)

- Below a critical size (~ 10 – 20 nm for $BaTiO_3$) ferroelectricity weakens and the cubic (paraelectric) phase is stabilised
- Nanocomposites of polymer + ceramic have high breakdown strength, so they are used in energy-storage capacitors, embedded capacitors and gate dielectrics (high-K: HfO_2)
- Dielectric loss tan $\delta = \varepsilon''/\varepsilon'$ depends on frequency; interfaces can raise loss

7.4 Thermal properties

- Lattice thermal conductivity $\kappa = (1/3)\, C\, v\, \lambda_{ph}$ (C = heat capacity, v = sound velocity, λ_{ph} = phonon mean free path)
- **Melting point falls** with size because surface atoms have fewer bonds: Au melts at 1064 °C in bulk but about 300 °C for 2 nm particles (Gibbs-Thomson effect)
- Phonon scattering at grain boundaries / surfaces reduces κ in nanostructures (Si nanowires have ~ 100 times smaller κ than bulk Si). This is great for thermoelectrics with figure of merit $ZT = S^2\sigma T/\kappa$
- Specific heat increases (more surface and grain-boundary atoms with extra vibrational modes); thermal expansion of nanocrystals is larger
- On the other hand, CNTs and graphene have extremely high thermal conductivity (3000 – 5000 W/mK), useful for heat spreaders
- Nanofluids (Al_2O_3, CuO or CNT in water) enhance heat transfer coefficients

7.5 Magnetic properties

Type	Behaviour	Susceptibility χ	Example
Diamagnetic	Weakly repelled, no permanent moment	small negative	Cu, Bi, water
Paramagnetic	Weakly attracted, random moments	small positive (Curie law)	Al, Pt, O_2

Ferromagnetic	Parallel aligned moments, domains, hysteresis, Curie temp T_C	large positive	Fe, Co, Ni
Antiferromagne	Neighbouring moments antiparallel	small positive; Neel temp T_N	MnO, Cr
Ferrimagnetic	Unequal antiparallel moments	large positive	Fe_3O_4, ferrites

- Ferromagnet breaks into domains to reduce magnetostatic energy. Below a critical single-domain size (~ 15 nm for Fe, 70 nm for Co) a particle becomes a single domain, so coercivity H_c is maximum
- Below that size thermal energy flips the moment: the particle is **superparamagnetic** (large moment, zero coercivity and remanence). Blocking temperature $T_B = KV / (25\ K_B)$ - where K = anisotropy constant, V = volume
- Surface atoms have broken exchange bonds, so saturation magnetisation often falls for very small particles. Antiferromagnetic nanoparticles can show weak ferromagnetism
- **Giant Magnetoresistance (GMR)**: multilayers of Fe/Cr or Co/Cu change resistance strongly with field (Fert and Gruenberg, Nobel 2007) - used in hard-disk read heads
- Applications: ferrofluids, MRI contrast agents, magnetic hyperthermia, high-density data storage, drug targeting

7.6 Opto-electronic properties

- Optoelectronics deals with devices that convert light into electricity or electricity into light: solar cells, LEDs, lasers, photodetectors
- **Quantum size effect**: as semiconductor size drops, the band gap widens, so absorption and emission shift to shorter wavelength (blue shift). CdSe dots change colour from red (large, ~ 6 nm) to blue (small, ~ 2 nm)
- **Surface Plasmon Resonance (SPR)**: collective oscillation of conduction electrons of Au / Ag nanoparticles in resonance with light. Au (~ 20 nm) absorbs near 520 nm, hence ruby-red colour; position shifts with size, shape and medium

- Exciton: electron-hole pair bound by Coulomb force with Bohr radius a_B. When particle radius < a_B the strong confinement regime occurs; exciton binding energy and oscillator strength rise
- Photoluminescence quantum yield of dots can exceed 90 %; sharp, size-tunable emission; large absorption cross-section
- Bulk Si is indirect and non-emitting, but porous / nano Si emits visible light

8. Effect of Size Reduction on Properties

Property	Change as size decreases
Surface area / volume	Increases enormously (S/V = 3/r); more reactive, better catalysts
Melting point	Decreases (Au 1064 °C to ~ 300 °C at 2 nm)
Lattice parameter	Usually contracts slightly due to surface stress
Hardness / strength	Increases (Hall-Petch) until about 10 nm
Band gap	Increases for semiconductors (blue shift)
Electrical resistivity	Increases for metals (surface scattering); conductance quantised
Magnetism	Single domain, then superparamagnetism; coercivity peaks
Optical	SPR colour in metals, size-dependent fluorescence in dots
Thermal conductivity	Decreases (phonon scattering); good for thermoelectrics
Dielectric	Space-charge polarisation increases; ferroelectricity vanishes under critical size

Summary:
Two root causes: (1) surface / interface atoms dominate; (2) quantum confinement of electrons, phonons and excitons.

9. Electronic Structure of Nanomaterials

Quantum confinement

If a dimension L is comparable with the de Broglie wavelength $\lambda = h / \sqrt{(2m^*E)}$ of the carriers, the electron is confined and its energy is quantised. For a particle in a 1D infinite well:

$$E_n = n^2 h^2 / (8 m^* L^2), \qquad n = 1, 2, 3 \ldots$$

- Energy $\propto 1/L^2$. Smaller L means bigger level spacing and a larger gap
- **Quantum well** (2D system): confined in 1 direction, free in 2. **Quantum wire** (1D): confined in 2, free in 1. **Quantum dot** (0D): confined in all 3 - "artificial atom"

Density of states g(E)

System	Free dim.	g(E) dependence	Shape
Bulk (3D)	3	$\propto E^{1/2}$	Smooth parabola
Quantum well (2D)	2	independent of E in each sub-band	Staircase
Quantum wire (1D)	1	$\propto E^{-1/2}$	Sharp spikes (van Hove singularities)
Quantum dot (0D)	0	$\delta(E - E_n)$	Discrete delta lines like an atom

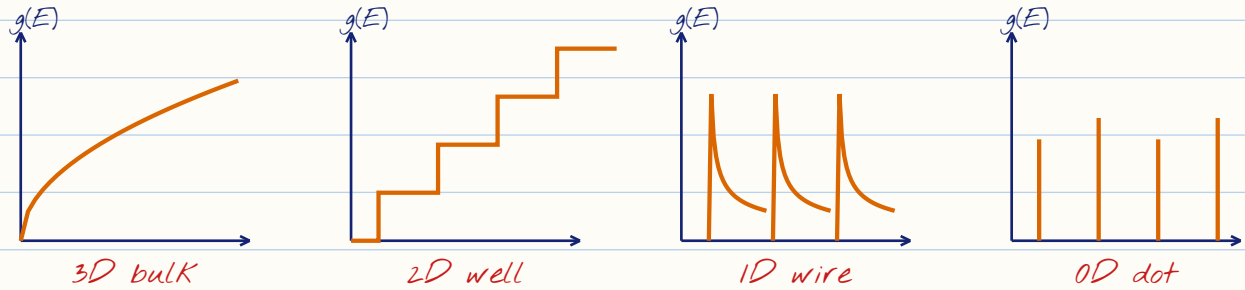


Fig: Density of states with decreasing dimensionality

Brus equation - size dependence of band gap

For a spherical semiconductor nanocrystal of radius R, the effective optical gap is:

$$E_g(R) = E_g(bulk) + \hbar^2\pi^2 / (2R^2)(1/m_e^* + 1/m_h^*) - 1.8\, e^2 / (4\pi\, \varepsilon_0\, \varepsilon_r\, R)$$

- 1st term: bulk band gap. 2nd term: confinement energy $\propto 1/R^2$ (blue shift). 3rd term: electron-hole Coulomb attraction $\propto 1/R$ (red shift, smaller effect)
- **Weak confinement**: $R > a_B$; **strong confinement**: $R < a_B$ (exciton Bohr radius $a_B = 4\pi\, \varepsilon_0\, \varepsilon_r\, \hbar^2 / (\mu\, e^2)$)
- Examples of a_B: CdS 3 nm, CdSe 5.6 nm, GaAs 12 nm, PbS 18 nm

Other consequences

- Discrete, atom-like levels give sharp absorption / emission peaks and "spectroscopic" size sorting
- Fermi level concept: for metal clusters, level spacing $\delta \approx 4E_F / 3N$ becomes larger than k_BT at small N: **Kubo gap**, so a metal cluster may become non-metallic below ~ 1 – 2 nm
- "Magic numbers" of atoms (e.g. 2, 8, 20, 40, 58 … for alkali clusters; 13, 55, 147 for icosahedral packing) indicate specially stable clusters, as in the jellium shell model

Quick Revision – Important Questions

1. Define nanometer, nanomaterial and nanotechnology. Why do properties change at the nano scale?
2. Classify nanomaterials with examples (0D, 1D, 2D, 3D).
3. Explain the seven crystal systems and 14 Bravais lattices.
4. How are directions and planes represented? Find the Miller indices of a plane with intercepts 2a, 3b, c.
5. Describe the band structure of metals, semiconductors and insulators; distinguish direct and indirect gaps.
6. Write a note on mechanical and magnetic properties of nanomaterials (Hall-Petch, superparamagnetism).
7. Explain electronic structure of nanomaterials; derive the density of states for 3D, 2D, 1D, 0D.
8. Explain the effect of size reduction on optical, electrical and thermal properties.

UNIT - 2

SYNTHESIS AND FABRICATION

Bulk samples - Single crystals - Bottom-up - Top-down - Semiconductor nanostructures

1. Introduction - Two Approaches

Nanomaterials are made by two broad strategies:



Fig: Top-down vs bottom-up approach

	Top-down	Bottom-up
Principle	Bulk is reduced to nano size by mechanical / etching processes	Atoms / molecules / ions assemble into nano structures by chemical reactions
Examples	Ball milling, lithography, sputtering, etching, laser ablation	Sol-gel, hydrothermal, CVD, co-precipitation, self-assembly, MBE
Merits	Large-scale, good for patterns in electronics	Better size / shape control, fewer defects, cheaper, atomic precision
Demerits	Surface defects, contamination, wastage, costly equipment	Hard to scale and to position / interconnect nanoparticles

2. Synthesis of Bulk Polycrystalline Samples

A polycrystalline sample consists of many small randomly oriented crystallites (grains) separated by grain boundaries. It is the starting material for most ceramics, oxides, superconductors and thermoelectrics.

2.1 Solid-state reaction (ceramic method)

1. **Weighing**: high purity (99.9 %+) oxides / carbonates in stoichiometric ratio, e.g. $BaCO_3 + TiO_2$ for $BaTiO_3$.
2. **Mixing and grinding**: in agate mortar or ball mill with acetone / ethanol for 2 – 4 h so that the particles are intimately mixed (reaction occurs only at the contact points).
3. **Calcination**: heat at 0.6 – 0.8 T_m (e.g. 900 – 1100 °C) so that the carbonates decompose and the desired phase forms by inter-diffusion. $BaCO_3 + TiO_2 \rightarrow BaTiO_3 + CO_2$.
4. **Regrinding** and repeat calcination until a single phase is confirmed by XRD.
5. **Pelletising**: add 2 – 5 % PVA binder, press into discs in a die (~ 5 tonne) with a hydraulic press.
6. **Sintering**: heat pellet at 1200 – 1400 °C for hours; grains grow and pores close, densification up to 95 – 99 % of theoretical density.

> *Remember:*
> Merits: simple, cheap, bulk quantities. Demerits: high temperature, long time, inhomogeneity, large grains (micron size), volatile loss (Pb, Bi).

2.2 Other routes for polycrystals

- **Co-precipitation**: mix metal salt solutions and add precipitant (NaOH, NH_4OH) to precipitate hydroxides together; calcine to oxide. Gives atomic-level mixing and lower temperature
- **Sol-gel / Pechini / combustion synthesis**: citrate or glycine as fuel, self-propagating, produces fine powder in minutes
- **Arc melting / induction melting** for metals and alloys, intermetallics (under Ar atmosphere, remelting several times for homogeneity)
- **Hot pressing and Spark Plasma Sintering (SPS)**: pressure + pulsed current give dense nano-grained ceramics at lower T within minutes

3. Growth of Single Crystals

A single crystal has one continuous lattice without grain boundaries, needed for devices (Si wafers, lasers) and for studying anisotropic properties. Growth requires a supersaturated / supercooled state so nucleation and growth proceed in a controlled way.

3.1 Growth from melt

Used when the material melts congruently without decomposing.

(a) Bridgman - Stockbarger technique

- Material sealed in a pointed ampoule (quartz / graphite) is lowered slowly (1 - 10 mm/h) through a furnace temperature gradient from above T_m to below it
- Crystallisation starts at the tip (single nucleus, or seed) and proceeds upward
- Examples: GaAs, CdTe, CaF_2, LiF, alkali halides

(b) Czochralski (CZ) pulling method

- A seed crystal touches the melt in a crucible and is pulled upward (1 - 10 cm/h) while rotating (5 - 30 rpm); the crucible rotates in the opposite direction
- Melt temperature is held just above T_m; the crystal diameter is controlled by pull speed and temperature
- A neck (Dash necking) is first formed to eliminate dislocations; produces large, high-quality boules (up to 300 mm Si)
- Examples: Si, Ge, GaAs (LEC), sapphire, YAG, $LiNbO_3$

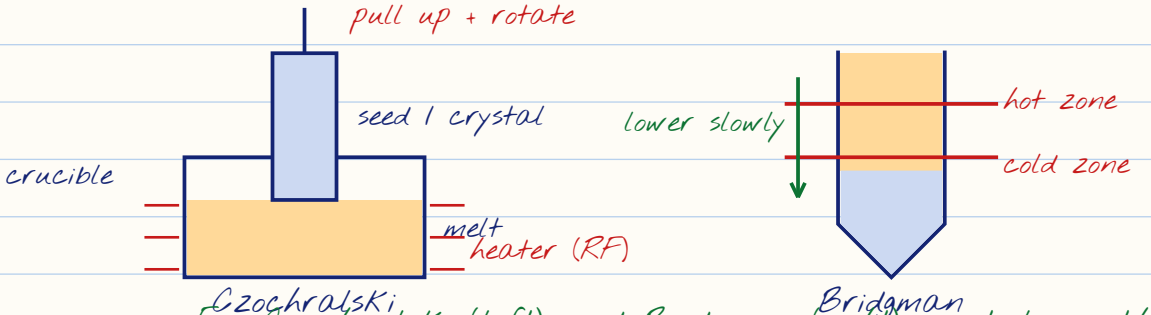


Fig: Czochralski (left) and Bridgman (right) crystal growth

(c) Other melt methods

- **Verneuil (flame fusion)**: powder falls through an oxy-hydrogen flame, melts and drops on a seed; synthetic ruby, sapphire, spinel

- **Zone melting / Float zone (FZ)**: a narrow molten zone travels along a rod; impurities segregate into the liquid (zone refining) so FZ-Si is ultra pure; no crucible contamination
- **Kyropoulos** method for large sapphire boules

3.2 Growth from solution

- **Low-temperature solution growth**: slow evaporation or slow cooling of a saturated solution (KDP, TGS, alum, NaCl); needs solubility with positive temperature coefficient, seed crystal in a constant temperature bath
- **Gel growth**: reactants diffuse slowly in silica gel; used for sparingly soluble compounds ($CaCO_3$, tartrates)
- **Flux growth**: material dissolved in a molten salt / oxide flux (PbO-PbF_2, Bi_2O_3) at high T and slowly cooled; for materials that melt incongruently (YIG, $BaTiO_3$)
- **Hydrothermal growth**: water at high T / P in an autoclave dissolves and re-crystallises material (quartz, ZnO) - see section 6

3.3 Growth from vapour

- **Physical vapour transport** (sealed tube sublimation), chemical vapour transport using a transport agent such as iodine: $MX + I_2 \rightleftharpoons MI_2 + X$, carried from hot to cold zone ($ZnSe$, TaS_2, MoS_2)
- **Epitaxial growth**: oriented overgrowth on a single crystal substrate (VPE, LPE, MBE, MOCVD)

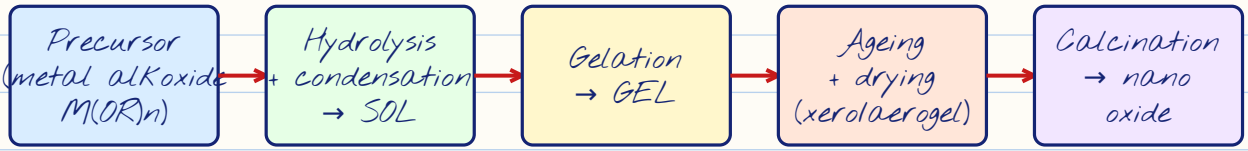
Method	Material suited	Advantages	Limitation
Czochralski	Si, Ge, GaAs, oxides	Large crystals, fast, high quality	Crucible contamination
Bridgman	Compounds, halides	Simple, volatile materials ok	Strain from the walls
Float zone	Si, refractory metals	Highest purity	Small diameter
Solution	Soluble salts, organics	Low T, low defect density	Slow (weeks)

Flux	Incongruent melting oxides	Growth below T_m	Flux inclusions
Vapour	Layered compounds, thin crystals	Pure, thin	Small size

4. Bottom-Up Approach

4.1 Sol-gel synthesis

A wet-chemical route in which a sol (colloidal particles dispersed in liquid) is converted into a gel (3D network containing liquid) and finally into a solid oxide.



H_2O, catalyst (acid / base), alcohol

Fig: Steps in sol-gel process

Reactions (e.g. TEOS / TTIP)

- **Hydrolysis**: $M\text{-}OR + H_2O \rightarrow M\text{-}OH + ROH$
- **Condensation (water)**: $M\text{-}OH + HO\text{-}M \rightarrow M\text{-}O\text{-}M + H_2O$
- **Condensation (alcohol)**: $M\text{-}OH + RO\text{-}M \rightarrow M\text{-}O\text{-}M + ROH$
- Final network of M-O-M bonds forms the gel. Example: $Si(OC_2H_5)_4 \rightarrow SiO_2$; $Ti(OC_3H_7)_4 \rightarrow TiO_2$
- **Effect of pH**: acid catalysis gives linear weakly branched chains (films, fibres); base catalysis gives highly branched dense particles (powders, Stober silica)
- Other factors: water-to-alkoxide ratio, temperature, solvent, precursor concentration
- **Drying**: ambient evaporation → xerogel (shrinks 5 - 10 times); supercritical drying → aerogel (very porous, density as low as 3 mg/cc, 99 % air, great insulator)
- **Coating techniques**: dip-coating, spin-coating (500 - 5000 rpm), spray; thickness ~ 50 - 500 nm per layer

Advantages	Disadvantages

Low temperature, high purity and homogeneity, control of particle size / composition, films, fibres, monoliths, powders, cheap equipment	Expensive alkoxides, shrinkage and cracking on drying, long processing time, organic residues, hard to avoid hydroxyl groups

Applications: antireflective and self-cleaning coatings, catalysts, sensors, optical fibres, bioactive glass, aerogel insulation, ceramic membranes.

4.2 Hydrothermal (solvothermal) synthesis

Crystallisation of materials from aqueous solution at <r>high temperature (100 – 300 °C) and pressure (1 – 100 MPa)</r> in a sealed Teflon-lined stainless steel autoclave. If a non-aqueous solvent is used, it is called solvothermal.

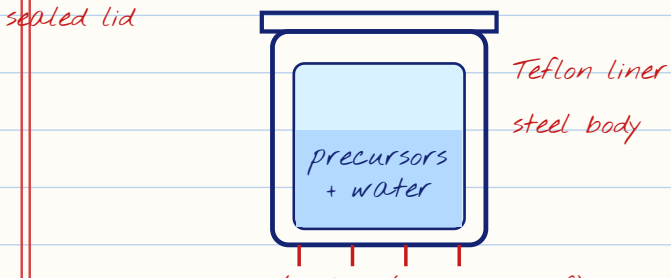


Fig: Teflon-lined hydrothermal autoclave

1. Dissolve salts, adjust pH
2. Fill 60 – 80 % of liner
3. Heat 12 – 48 h, autogenous pressure
4. Cool, wash, dry the powder

- Under these conditions water has lower viscosity and dielectric constant, so solubility of oxides increases; the dissolved species nucleate and grow when supersaturated
- Size and shape (rods, wires, flowers, cubes) are tuned by temperature, time, pH, concentration and surfactants / capping agents (CTAB, PVP)
- Examples: ZnO nanorods, TiO_2 nanotubes, $BaTiO_3$, Fe_3O_4, quartz single crystals (growth at 400 °C, 1 – 2 Kbar), zeolites, MoS_2
- **Merits**: one-step, crystalline product without calcination, low T, uniform morphology, closed system (no volatile loss). **Demerits**: costly autoclaves, cannot observe the reaction, safety at high pressure, long time
- **Microwave-hydrothermal** heating reduces the reaction time to minutes

5. Thin Film Growth

A thin film is a layer of material with thickness from a few nm to about 1 μm deposited on a substrate. Properties depend on thickness, substrate, microstructure and deposition conditions.

Growth modes (nucleation and growth)

- **Frank-van der Merwe** (layer-by-layer, 2D): adatom-substrate bonding stronger than adatom-adatom, gives smooth epitaxial films
- **Volmer-Weber** (island, 3D): adatom-adatom bonding stronger, islands coalesce into a film, metals on insulators
- **Stranski-Krastanov**: layer-by-layer for a few monolayers, then islands due to strain (e.g. InAs on GaAs gives self-assembled quantum dots)

Steps: adsorption → surface diffusion → nucleation → island growth → coalescence → continuous film. Epitaxy = oriented growth, homo-epitaxy (Si on Si) or hetero-epitaxy (GaN on sapphire); lattice mismatch $f = (a_s - a_f)/a_f$ controls strain and dislocations.

Physical (PVD)	Chemical (CVD / solution)
Thermal / e-beam evaporation, sputtering, MBE, pulsed laser deposition (PLD), arc	CVD, MOCVD, PECVD, ALD, spray pyrolysis, spin / dip coating, electrodeposition, CBD, SILAR

5.1 Physical Vapour Deposition (PVD)

Material is converted into vapour by a physical means in high vacuum (10^{-5} - 10^{-7} torr), transported and condensed on the substrate without chemical reaction. Vacuum gives long mean free path and clean surface.

(a) Thermal / vacuum evaporation

- Source (in W / Mo boat or filament, resistive heating) heated until vapour pressure is high (~ 10^{-2} torr); atoms travel in a straight line (line of sight) to the substrate
- **E-beam evaporation**: focused 5 - 10 KeV electron beam heats a water-cooled crucible, for high melting materials (Ti, SiO_2, W)
- Deposition rate 1 - 10 nm/s, monitored by a quartz crystal

thickness monitor (frequency shift)

- Limits: alloy composition changes (different vapour pressures), poor step coverage

(b) Sputtering

- Argon plasma (10^{-2} – 10^{-3} torr) is created between a target (cathode) and substrate (anode). Ar^+ ions are accelerated into the target and knock out atoms by momentum transfer; atoms condense on the substrate
- **DC sputtering** for metals; **RF sputtering** (13.56 MHz) for insulators / oxides; **Magnetron sputtering** uses magnets to trap electrons near the target, which raises ionisation and rate; **Reactive sputtering** with O_2 / N_2 to deposit oxides, nitrides (TiN, ZnO)
- Sputter yield depends on ion energy, mass and target; good adhesion and composition transfer; uniform large area films

(c) Pulsed Laser Deposition (PLD)

- Pulsed excimer laser (KrF 248 nm, 20 ns) is focused on a rotating target, creating an ablation plume that expands to the heated substrate in an oxygen background
- Stoichiometric transfer of complex oxides (YBCO, PZT, LSMO); good for epitaxial multi-layers; limitation: small area, droplets (particulates)

(d) Molecular Beam Epitaxy (MBE)

- Ultra high vacuum (10^{-10} torr). Effusion cells (Knudsen) produce atomic / molecular beams that reach the heated substrate without collisions; shutters switch the beams in < 1 s
- Growth rate ~ 1 monolayer/s (0.1 – 1 μm/h); **RHEED** shows oscillations giving in-situ monolayer control
- Used to make quantum wells, superlattices, quantum dots (GaAs / AlGaAs) and HEMT lasers. Very expensive and slow

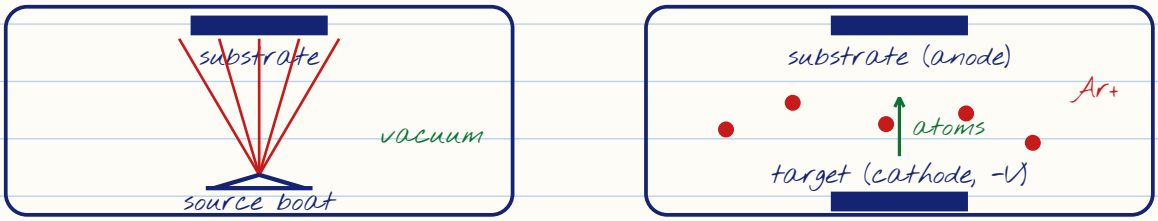


Fig: Schematic of evaporation and sputtering systems

5.2 Chemical Vapour Deposition (CVD)

A film is deposited by a chemical reaction of gaseous precursors on or near a heated substrate, volatile by-products are pumped away.

1. Precursor gases are transported into the reactor by a carrier gas (Ar / H_2).
2. They diffuse through the boundary layer to the hot substrate surface.
3. Adsorption → surface reaction / decomposition (e.g. $SiH_4 \rightarrow Si + 2H_2$ at 600 - 1000 °C).
4. Nucleation and film growth; by-products desorb and are exhausted.

Type	Key feature	Typical use
APCVD / LPCVD	Atmospheric / low pressure (0.1 - 10 torr) gives better uniformity	Poly-Si, SiO_2, Si_3N_4
PECVD	Plasma activates gas, so deposition at 200 - 400 °C	a-Si:H solar cell, passivation, DLC
MOCVD / MOVPE	Metal-organic precursors (TMGa, TMAl, NH_3)	GaN LEDs, laser diodes, solar cells
Thermal CVD	Hot wall / cold wall furnaces	CNT, graphene on Cu, diamond films
ALD	Self-limiting alternating pulses, one atomic layer per cycle	High-K Al_2O_3, HfO_2 gate oxide, conformal nanofilms

- **Advantages of CVD**: conformal coverage, high deposition rate, dense and pure, can coat 3D shapes, good for CNT / graphene / diamond growth

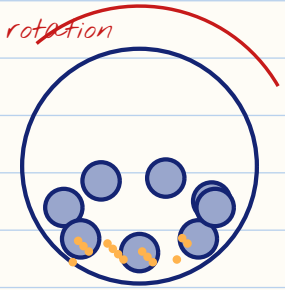
- **Disadvantages**: high temperature, toxic / flammable precursors, corrosive by-products
- **VLS growth** (Vapour-Liquid-Solid) of nanowires: Au catalyst droplet absorbs vapour (SiH_4) and supersaturates, the wire grows from the droplet interface. Wire diameter ≈ catalyst size

PVD	CVD
Physical process, no reaction	Chemical reaction at the surface
High vacuum, line of sight	Low to atmospheric pressure, conformal
Lower substrate temperature	Higher temperature (except PECVD)
Metals, alloys, simple compounds	Semiconductors, dielectrics, 2D materials

6. Top-Down Approach

6.1 Ball milling (mechanical attrition)

- A hardened steel / WC / zirconia vial is filled with balls (ball-to-powder weight ratio 10:1) and the powder; the vial rotates / vibrates at high speed (planetary mill: 200 – 600 rpm)
- Balls collide with the powder: repeated fracture, cold welding and re-fracture. After tens of hours grain size decreases to 5 – 20 nm
- Critical speed $N_c = 42.3/\sqrt{D}$ (rpm, D = mill diameter in m); operate at 65 – 75 % of N_c so that balls cascade (too slow no impact, too fast balls stick to the wall)
- **Mechanical alloying**: milling elemental powders to form alloys, amorphous phases, ODS superalloys, nanocomposites. Cryo-milling reduces recovery
- Process variables: milling time, speed, ball size / material, BPR, atmosphere (Ar), process control agent (stearic acid, methanol)



Ball mill vial

coarse powder → impact & shear
→ fracture / cold weld / re-fracture
→ nanocrystalline powder (5 – 20 nm)

Types: planetary, SPEX shaker, attritor

Fig: Ball milling

- **Merits**: simple, scalable, any material, low cost, dry process.
  Demerits: contamination from balls / vial, broad size distribution, agglomeration, strain and amorphisation

6.2 Micro fabrication

Set of techniques (from the IC industry) to create micron / nanometre patterned structures on a wafer: clean room (class 100 – 1000) → oxidation → deposition → lithography → etching → doping (diffusion / ion implantation) → metallisation → packaging.

- **Substrate preparation**: Si wafer cleaning (RCA clean: H_2O_2 + NH_4OH / HCl), thermal oxidation SiO_2 growth (dry / wet) as mask or insulator
- **Etching** removes unwanted material: **wet etching** (KOH, HF; isotropic or anisotropic along crystal planes) and **dry etching** (plasma RIE, DRIE Bosch process; anisotropic, vertical walls)
- **Doping** by ion implantation (energy 10 – 200 KeV, dose 10^{11} – 10^{16} cm^{-2}) + annealing
- **Bulk and surface micromachining**, LIGA, make MEMS / NEMS: cantilevers, accelerometers, micro-mirrors, microfluidic chips
- **Lift-off**: metal deposited over patterned resist, resist dissolves, leaving the metal only in openings

6.3 Lithography

Lithography (Greek: stone writing) transfers a pattern from a mask / design to a radiation-sensitive layer (resist) on a substrate.

1. **Substrate cleaning and priming** (HMDS for adhesion).
2. **Spin-coating the resist** (e.g. 1 μm at 3000 rpm), then soft bake (90 – 110 °C).
3. **Exposure** through a mask (or direct writing) to UV / e-beam /

X-ray.

4. **Development**: positive resist - exposed area dissolves; negative resist - exposed area cross-links and remains.
5. **Pattern transfer**: etch or deposit (lift-off); finally strip the resist.

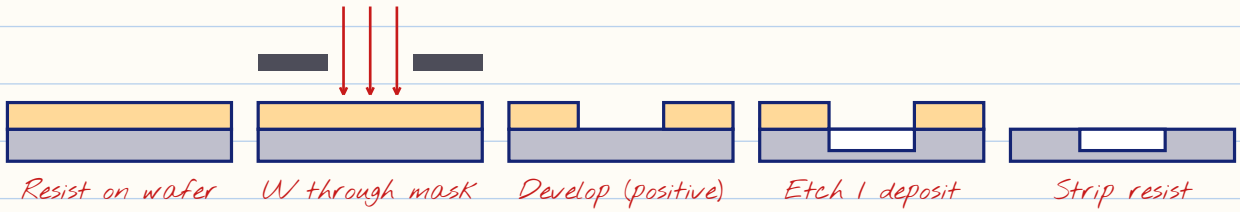


Fig: Photolithography steps

Type	Source	Resolution	Remarks
Photolithography	UV, DUV (248 nm KrF, 193 nm ArF), EUV 13.5 nm	down to ~ 5 - 7 nm node with EUV + multipatterning	Mass production, parallel, mask needed; diffraction limit $R = k_1 \lambda / NA$
Electron beam (EBL)	Focused electron beam, PMMA resist	~ 5 - 10 nm	Maskless, serial, slow, expensive; for masks and research
X-ray lithography	0.4 - 5 nm X-rays	< 50 nm	Large depth of focus, mask difficult, LIGA
Ion beam (FIB)	Ga^+ ions	~ 10 nm	Direct milling and deposition, TEM sample prep
Nano-imprint (NIL)	Mold pressed on resist, UV / thermal cure	< 10 nm	Cheap, high throughput, mold wear
Soft lithography / Dip-pen	PDMS stamp; AFM tip with ink	~ 30 - 100 nm	Printing of biomolecules, self-assembled monolayers

7. Requirements for Realising Semiconductor Nanostructures

To make quantum wells, wires and dots that really show quantum effects, the following requirements must be satisfied:

1. **Size / dimension control**: confinement length must be ≤ de Broglie wavelength or exciton Bohr radius (~ 5 – 20 nm), with size dispersion < 5 %, otherwise optical lines broaden (inhomogeneous broadening).
2. **High crystalline quality and interfaces**: atomically sharp, defect free interfaces; lattice matching (GaAs/AlGaAs $a \approx 5.65$ Å) or controlled strain; low density of dislocations and traps (which kill luminescence).
3. **Band gap engineering**: choose a pair of materials with suitable band offsets (ΔE_c, ΔE_v) so that carriers are confined in the narrow-gap material (type-I heterostructure).
4. **Purity and doping control**: ultrahigh vacuum, high-purity sources; **modulation doping** (dopants in the barrier) gives a 2D electron gas with very high mobility (> 10^7 cm^2/Vs).
5. **Surface passivation**: surface states must be removed by capping / shell (CdSe/ZnS core-shell, organic ligands) otherwise carriers recombine non-radiatively.
6. **Positioning, ordering and reproducibility** so devices can be integrated; low temperature measurement (carrier thermal energy $K_BT \ll$ level spacing).
7. **Stability** against oxidation and inter-diffusion at working temperature.

Structure	Common materials	Typical preparation
Quantum well / superlattice	GaAs/AlGaAs, InGaAs/InP, Si/SiGe	MBE, MOCVD (monolayer control)
Quantum wire	InP, Si, GaAs, ZnO nanowires	VLS CVD, MBE on V-grooves, e-beam lithography + etching
Quantum dot	CdSe, PbS, InAs, Si, InP, perovskite	Colloidal chemistry (hot injection), Stranski-Krastanov MBE / MOCVD, lithography

8. Growth Techniques for Nanostructures

8.1 Epitaxial growth methods

- **MBE** - UHV, atomic beams, highest control (monolayer). Quantum wells, superlattices, self-assembled QDs
- **MOCVD / MOVPE** - metal-organics + hydrides at 500 - 800 °C, fast and scalable, GaN LEDs, laser diodes, fewer cost per wafer
- **Liquid phase epitaxy (LPE)** - film grows from a supersaturated melt on a substrate, thick films, low cost, simple
- **Chemical beam epitaxy (CBE)** - metal organic beams in UHV (hybrid)
- **Atomic layer epitaxy / ALD** - one atomic layer per cycle

8.2 Self-assembled quantum dots (Stranski-Krastanov)

When a lattice-mismatched film (InAs on GaAs, mismatch 7 %) grows beyond a critical thickness (~ 1.7 monolayer) strain relaxes by forming coherent dislocation-free islands on a thin wetting layer. Dots of 10 - 30 nm base, 3 - 10 nm height with density 10^{10} - 10^{11} cm^{-2} are obtained and can be capped by GaAs.

8.3 Vapour-Liquid-Solid (VLS) for nanowires

Metal catalyst (Au) forms a liquid eutectic alloy droplet with the semiconductor (Au-Si eutectic at 363 °C). Precursor vapour dissolves in the droplet, on supersaturation, solid precipitates under the droplet, extending a wire. Diameter is controlled by the droplet size, also vapour-solid (VS) and solution-liquid-solid (SLS) variants.

8.4 Colloidal synthesis of quantum dots

- **Hot injection** (Murray, Bawendi): a cold solution of precursor (e.g. TOP-Se) is injected into a hot (250 - 300 °C) solution of Cd precursor with coordinating ligands (TOPO, oleic acid). A burst of nucleation is followed by slow growth at a lower temperature (LaMer model)
- Size controlled by time and temperature, ligands (capping agents) stop aggregation
- **Core-shell** (CdSe/ZnS) to improve photoluminescence yield and

stability

- **Ostwald ripening**: larger particles grow at the cost of smaller ones; narrow size distribution requires nucleation and growth to be separated

> Summary:
> Top-down gives patterned structures of 10s of nm but with edge damage; bottom-up epitaxy and colloidal chemistry give atomically precise, defect-free structures. Most devices combine both.

Quick Revision - Important Questions

1. Describe solid-state reaction method for the synthesis of polycrystalline sample.
2. Explain Czochralski and Bridgman methods of single crystal growth with neat diagrams.
3. Explain the sol-gel process: steps, chemistry, advantages and applications.
4. Describe hydrothermal synthesis with a diagram.
5. Distinguish PVD and CVD. Explain sputtering / evaporation and MOCVD.
6. Explain the principle of ball milling and its merits / demerits.
7. Describe photolithography and e-beam lithography.
8. What are the requirements for realising semiconductor nanostructures? Discuss MBE and VLS growth.

UNIT - 3

CHARACTERIZATION TECHNIQUES

XRD - SEM - TEM - SPM - AFM - PFM - XPS - XAS - ARPES - DRS - PL - Raman

1. Overview

Characterization reveals the <r>structure, size, shape, composition, surface and physical properties</r> of nanomaterials. Probes used are X-rays, electrons, photons or a sharp physical tip.

Technique	Probe	Information
XRD	X-rays	Crystal structure, phase, lattice parameter, crystallite size, strain
SEM	Electrons (reflect)	Surface morphology, topography, composition (EDX)
TEM	Electrons (transmit)	Internal structure, size, lattice fringes, diffraction (SAED)
STM / AFM / PFM	Sharp tip	3D surface image, atomic resolution, forces, piezo / ferroelectric domains
XPS	X-rays → photoelectrons	Surface composition, oxidation state, chemical bonding
XANES / EXAFS	Tunable X-rays (synchrotron)	Oxidation state, local coordination, bond lengths
ARPES	UV / X-ray → photoelectrons	Band structure E(K), Fermi surface
DRS (UV-Vis)	UV - visible light	Optical absorption, band gap
PL	Light → emission	Defects, recombination, emission energy, quantum yield
Raman	Laser → inelastic scattering	Vibrational modes, phases, defects, graphene layers, CNT

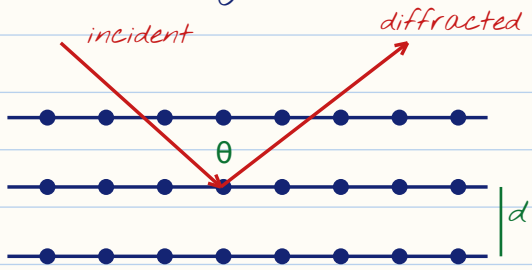
2. X-Ray Diffraction (XRD) and Scherrer Method

X-rays ($\lambda \sim 0.5 - 2.5$ Å, e.g. Cu $K\alpha = 1.5406$ Å) have wavelength comparable to inter-atomic spacing, so a crystal acts as a 3D diffraction grating.

Bragg's law

$$2\, d_{hkl} \sin\theta = n\lambda$$

Constructive interference occurs when the path difference between rays scattered from adjacent planes ($2d \sin\theta$) equals an integer multiple of λ. Here d = interplanar spacing, θ = glancing angle (diffraction angle is 2θ).



Source → sample → detector

Detector moves at 2θ while the sample rotates at θ ($\theta - 2\theta$ geometry)

Output: intensity vs 2θ plot

Fig: Bragg diffraction from crystal planes

Working

- Cu target X-ray tube (40 KV, 30 mA) → Ni filter removes $K\beta$ → monochromatic $K\alpha$ beam falls on the powdered sample on a holder
- Goniometer scans 2θ (typically 10° - 90°, step 0.02°), a scintillation / solid state detector records the intensity
- Each crystalline phase gives a unique set of peaks (fingerprint). Compare with JCPDS / ICDD database cards for phase identification
- Peak position gives d, so lattice parameter. For cubic: $a = d\sqrt{(h^2 + k^2 + l^2)}$
- Peak intensity gives atomic positions / texture; peak width gives crystallite size and strain; amorphous phase gives a broad hump

Scherrer method - crystallite size

Finite crystal size broadens the XRD peaks (in a perfect infinite

crystal peaks are sharp). Scherrer related the mean crystallite size D to the broadening:

$$D = K \lambda / (\beta \cos \theta)$$

- K = shape factor (≈ 0.9 for spherical particles; 0.89 – 1.0), λ = X-ray wavelength, θ = Bragg angle of the peak
- β = FWHM (full width at half maximum) of the peak in **radians** after subtracting the instrumental broadening: $\beta^2 = \beta_{obs}^2 - \beta_{inst}^2$
- Valid for D < ~ 100 nm. Gives the size of coherently diffracting domains, which may be smaller than the particle size

Worked example:

Example: peak at 2θ = 38.2°, FWHM = 0.4° = 0.00698 rad, λ = 1.5406 Å, K = 0.9. D = 0.9 × 1.5406 / (0.00698 × cos 19.1°) = 1.3865 / 0.00659 ≈ 210 Å = 21 nm.

Williamson-Hall analysis (size + strain)

$$\beta \cos \theta = K \lambda / D + 4 \varepsilon \sin \theta$$

Plot β cosθ (y) against 4 sinθ (x): the intercept gives size D, the slope gives micro-strain ε. Other use: dislocation density $\delta = 1/D^2$.

- Advantages of XRD: non-destructive, quick, quantitative phase analysis, needs only mg of sample. Limitation: needs crystalline sample, poor for amorphous, weak for light elements; gives average (bulk) information
- Related: SAXS (small angle) gives particle size 1 – 100 nm; thin film XRD (grazing incidence, GIXRD), rocking curve for epitaxial quality; Rietveld refinement for structure

3. Scanning Electron Microscopy (SEM)

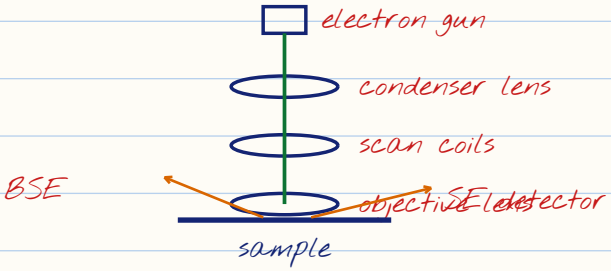
SEM forms an image of the surface by scanning a finely focused electron beam over the sample and collecting the emitted signals. Resolution ~ 1 – 10 nm, magnification 10 – 500,000 ×, large depth of field (3D-like image).

Principle and components

1. **Electron gun** (tungsten filament, LaB_6 or field emission gun FEG)

produces electrons accelerated by 1 - 30 KV.

2. **Condenser lenses** (electromagnetic) demagnify the beam to a spot of 1 - 10 nm; **objective lens** focuses it on the sample.
3. **Scan coils** deflect the beam in a raster pattern over the surface.
4. **Detectors** collect the signals; the display is synchronised with the scan, brightness at each pixel = signal intensity.
5. **High vacuum** (10^{-4} - 10^{-6} Pa) chamber prevents scattering of electrons and contamination.



Signals from the sample:

Secondary electrons (SE, < 50 eV) → topogra
Back-scattered electrons (BSE) → Z contra
Characteristic X-rays → EDX (elemental a
Auger electrons → surface composition
Cathodoluminescence → optical

Fig: Schematic of a scanning electron microscope

- **Secondary electrons** come from the top few nm and give high resolution topographic contrast (edges look bright).
  Back-scattered electrons are high energy electrons reflected by nuclei; the yield increases with atomic number Z so heavy-element regions look bright (composition contrast)
- **EDX / EDS**: the energy of characteristic X-rays identifies elements (qualitative and quantitative, atomic %, mapping)
- **Sample preparation**: dry, vacuum-stable, non-conducting samples are coated with 5 - 10 nm Au / Pt / C by sputtering to avoid charging, mounted on carbon tape / stub. ESEM allows wet samples
- **Uses**: particle shape and size distribution, nanowire length, film thickness (cross-section), grain structure, fracture surfaces, porosity
- Limitation: only surface, 2D picture of a 3D surface, no crystal-structure information, resolution lower than TEM

4. Transmission Electron Microscopy (TEM)

A high energy electron beam (80 - 300 KV) passes through an

ultra-thin sample (< 100 nm), transmitted and diffracted electrons form an image. Resolution < 0.1 nm (atomic), magnification up to 10^6 - 10^7 x.

$$\lambda = h / \sqrt{(2 m e V (1 + eV / 2 m c^2))} \approx 1.226 / \sqrt{V} \text{ nm}$$

At 200 KV, λ = 2.5 pm, far less than the wavelength of light, so atomic resolution is possible.

Working

1. Electron gun (W, LaB_6, FEG) and a stack of condenser lenses produce a parallel beam on the specimen.
2. The specimen (on a 3 mm Cu grid with carbon film) scatters electrons by thickness, density, Z and crystal orientation.
3. The objective lens forms the first image / diffraction pattern; intermediate and projector lenses magnify it onto a fluorescent screen or CCD / CMOS camera.
4. Entire column under high vacuum 10^{-5} - 10^{-7} Pa.

Mode	What is seen	Use
Bright field	Image from direct beam; dense regions dark	Morphology, size distribution
Dark field	Image from a diffracted beam, that crystal bright	Identify particular phases / defects
HRTEM	Phase contrast, lattice fringes	Atomic planes, d-spacing, defects, core-shell
SAED	Diffraction pattern (rings: polycrystal, spots: single crystal)	Crystal structure, orientation
STEM-HAAI	Focused scanning beam, Z-contrast	Atomic columns, EELS / EDX mapping

- Sample preparation: nanoparticles are dispersed in ethanol, sonicated, drop-cast on a carbon-coated grid. Bulk is thinned by mechanical polishing + ion milling, ultramicrotomy or FIB
- Merits: highest resolution, imaging + diffraction + chemistry (EELS) together. Demerits: costly, thin sample needed, electron beam

damage, small area sampled

SEM	TEM
Surface imaging by reflected / emitted electrons	Internal structure by transmitted electrons
1 - 30 KV; resolution ~ 1 - 10 nm	80 - 300 KV; resolution < 0.1 nm
Bulk samples, little preparation	Ultra-thin sample (< 100 nm)
3D-like topography, large depth of field	2D projection image
Cheaper, easier	Expensive, skilled operation

5. Scanning Probe Microscopy (SPM)

A family of microscopes in which a <r>sharp probe (tip, radius 1 - 50 nm) is scanned very close to the surface</r> and a local tip-sample interaction is recorded to build an image. It does not use lenses, so resolution is set by tip sharpness and distance, not by wavelength. It works in air, liquid or vacuum.

Scanning Tunnelling Microscopy (STM) - Binnig and Rohrer 1981

- A metallic tip (W / Pt-Ir) is brought within ~ 0.5 - 1 nm of a conducting surface, a bias voltage (mV - V) causes electrons to **quantum mechanically tunnel** across the vacuum gap
- Tunnelling current: $I \propto V \exp(-2\kappa d)$ with $\kappa = \sqrt{(2m\varphi)}/\hbar$. Current changes by a factor of ~ 10 for each 0.1 nm change in gap (extreme sensitivity), giving vertical resolution of 0.01 nm and lateral of 0.1 nm: individual atoms are imaged
- **Constant current mode**: feedback adjusts the tip height (piezo-scanner) to keep I constant, z-signal = topography. **Constant height mode**: fast, records I
- Limitation: sample must be conducting / semiconducting; STS (spectroscopy) measures the local density of states

Common features of SPM

- Piezoelectric scanner (tube) gives sub-Å movement in x, y, z
- Feedback loop (PID) controls the tip-surface interaction

- Vibration isolation is essential, image formed pixel by pixel line by line
- Variants: AFM (force), MFM (magnetic force), EFM (electrostatic), KPFM (surface potential), SNOM / NSOM (near-field optical), SThM (thermal), PFM (piezo response), SECM, conductive AFM

6. Atomic Force Microscopy (AFM)

Invented by Binnig, Quate and Gerber (1986). Measures the <r>force between a sharp tip and the surface</r>, works on insulators as well, so it is more versatile than STM.

Construction and working

- **Cantilever** (Si or Si_3N_4, 100 – 200 μm long, spring constant 0.1 – 50 N/m) with an integrated sharp tip (radius 5 – 20 nm)
- **Laser and photodiode detector**: a laser beam reflects off the back of the cantilever onto a four-quadrant photodiode; tiny deflection is magnified into a large spot movement (optical lever)
- **Piezo scanner** moves the sample (or tip) in the x-y raster; **feedback loop** keeps the deflection or oscillation amplitude constant and the z voltage is recorded as the height image
- Force obeys Hooke's law: $F = -k\,z$. Forces: van der Waals (attractive at long range), Pauli repulsion (short range), capillary, electrostatic, magnetic

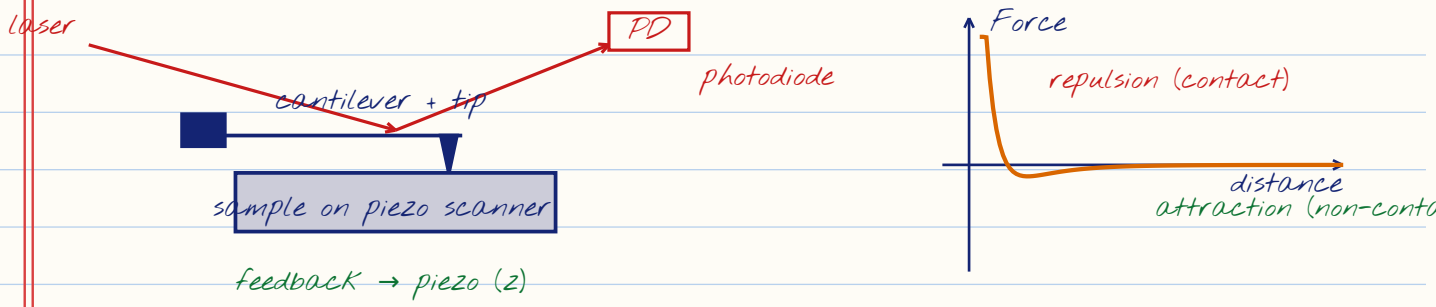


Fig: AFM principle (optical lever) and force-distance curve

Mode	Tip-sample	Features
Contact	Tip touches surface, repulsive force	Fast, high resolution; may damage soft samples, lateral forces

Non-contact	Tip 1 – 10 nm above, attractive vdW force, cantilever oscillates near resonance	No wear; lower resolution, can be affected by water layer
Tapping (intermittent)	Oscillating cantilever lightly taps the surface, amplitude is the feedback signal; phase image shows material contrast	Most popular; minimal damage; suits polymers, DNA, soft nanoparticles

- Information: 3D topography, roughness (Ra, RMS), particle / grain size and height, step height of films (sub-nm), mechanical (nano-indentation, modulus), adhesion, friction (LFM), conductivity (c-AFM), magnetic domains (MFM), surface potential (KPFM)
- **Advantages**: no vacuum or conducting sample needed, can work in liquid (biological samples), true 3D height data. **Disadvantages**: small scan area (~ 100 μm), slow, tip artefacts (convolution), image drift

STM	AFM
Measures tunnelling current	Measures force / cantilever deflection
Conducting samples only	Conducting and insulating samples
Needs vacuum (usually)	Air, liquid or vacuum
Atomic resolution routine	Atomic resolution in non-contact UHV; typically nm

7. Piezoresponse Force Microscopy (PFM)

PFM is an AFM-based (contact mode) technique used to image and manipulate ferroelectric and piezoelectric domains at the nanoscale. It uses the converse piezoelectric effect: a material deforms when an electric field is applied.

- A conductive tip (Pt / Ir-coated) in contact with the sample acts as the top electrode. An AC voltage $V = V_0 \cos \omega t$ (+ DC) is

applied between tip and the bottom electrode

- The sample expands / contracts: $\Delta z = d_{33} V_0 \cos(\omega t + \varphi)$. The cantilever follows this tiny vibration (pm range) and a lock-in amplifier extracts the amplitude and phase
- **Amplitude** → magnitude of piezoelectric coefficient d_{33} (domain strength; zero at domain walls). **Phase** → polarisation direction: in-phase (0°) for P pointing down, out-of-phase (180°) for P pointing up
- **Vertical PFM** detects out-of-plane polarisation; **lateral PFM** (torsion of the cantilever) detects in-plane polarisation
- **Local polarisation switching**: sweeping a DC bias gives piezoresponse hysteresis loops (coercive voltage, imprint), one can **write domains** by applying a bias above the coercive voltage
- **Applications**: ferroelectric thin films (PZT, $BiFeO_3$, $BaTiO_3$), nanostructured ferroelectrics for FeRAM, multiferroics, piezo nanowires (ZnO), biological piezoelectricity (collagen)
- **Limitation**: electrostatic artefacts, tip wear, needs conductive tip; Dual-AC resonance tracking (DART) improves signal

8. X-ray Photoelectron Spectroscopy (XPS / ESCA)

XPS is a surface sensitive technique (top 1 - 10 nm) that identifies the elements, their chemical state and composition from the photoelectrons emitted when the sample is irradiated by soft X-rays (Al Kα 1486.6 eV or Mg Kα 1253.6 eV). It is based on the photoelectric effect.

$$E_B = h\nu - E_K - \varphi_{spec}$$

- E_B = binding energy of the core level; E_K = measured kinetic energy; φ = spectrometer work function. E_B is a signature of each element and orbital (1s, $2p_{3/2}$ ...)
- **Setup**: X-ray source + monochromator → sample in UHV (10^{-9} torr) → electron energy analyser (hemispherical) → detector. UHV needed because electrons are scattered by gas molecules and the surface would be contaminated
- **Survey scan** (0 - 1100 eV) identifies all elements except H and He;

high-resolution scan of a peak gives chemical state

- **Chemical shift**: change in E_B (0.1 – 10 eV) with oxidation state / bonding; e.g. C 1s: C–C 284.8 eV, C–O 286.5, C=O 288, O–C=O 289 eV; Ti $2p_{3/2}$ 458.8 eV is Ti^{4+}; Fe^{2+} / Fe^{3+} distinguished
- **Spin-orbit splitting**: p, d, f levels split into doublets ($2p_{3/2}/2p_{1/2}$ area ratio 2:1, d 3:2, f 4:3)
- **Quantification**: atomic % ∝ peak area / sensitivity factor. Depth profile by Ar^+ ion sputtering; **angle-resolved XPS** for thin film thickness; XPS mapping; also valence band edge
- Charge referencing with adventitious C 1s = 284.8 eV (insulating samples); other features: shake-up satellites, Auger lines
- Uses in nano: surface oxidation, doping, core-shell composition, functionalisation of CNT / graphene (C–O, C–N), catalysts, thin film interfaces
- Limitations: surface only, UHV, cannot detect H / He, expensive, X-ray damage to some samples

9. XANES and XAFS (X-ray Absorption Spectroscopy)

When X-ray energy is scanned across the absorption edge of an element, the absorption coefficient $\mu(E)$ rises sharply (edge) as a core electron is ejected, followed by oscillations. Needs a **synchrotron** source with a tunable, intense X-ray beam. It probes the <r>local atomic and electronic structure</r> of a chosen element, even in amorphous materials, solutions or very dilute nano systems.

Region	Energy range	Information
Pre-edge	Few eV before edge	Forbidden 1s → 3d transitions; coordination symmetry (tetrahedral vs octahedral)
XANES (NEXAFS)	Edge to ~ 50 eV above	Oxidation state (edge shifts to higher E with higher valence), coordination geometry, unoccupied DOS, covalency

EXAFS	~ 50 – 1000 eV above edge	Bond distance, coordination number, disorder (Debye-Waller factor) of neighbouring shells

XAFS = XANES + EXAFS. In EXAFS the outgoing photoelectron wave scatters off neighbouring atoms and interferes with itself, modulating $\mu(E)$:

$$\chi(k) = \sum_j N_j \, S_0^2 \, f_j(k) \,/\, (k \, R_j^2) \cdot \sin(2kR_j + \delta_j) \cdot e^{-2\sigma_j^2 k^2} \, e^{-2R_j/\lambda}$$

- Fourier transform of $k^3\chi(k)$ gives a radial distribution function with peaks at the neighbour distances R_j
- Modes: transmission (concentrated samples), fluorescence yield (dilute / thin film), electron yield (surface). Edges: K, L_{III}
- Uses: oxidation state and site of dopants in nano oxides (Mn in ZnO), size-dependent bond contraction, active sites in nanocatalysts, in-situ battery studies
- Advantage: element specific, no long-range order needed. Limitation: needs synchrotron, data analysis is complex (Athena / Artemis)

10. Angle Resolved Photoemission Spectroscopy (ARPES)

ARPES is the direct experimental probe of the <r>electronic band structure $E(k)$<l/r>: by measuring the kinetic energy and emission angle of photoelectrons ejected by UV / soft X-ray photons (He I 21.2 eV, synchrotron 20 – 100 eV, lasers 6 eV).

- Energy conservation: $E_B = h\nu - \phi - E_K$. Momentum parallel to the surface is conserved when the electron leaves the crystal:

$$\hbar k_{\parallel} = \sqrt{2 m E_K} \sin\theta$$

- A hemispherical analyser with a 2D detector records intensity vs E_K and angle θ in one shot, so E_B vs $k_{\parallel}$ (band dispersion) is mapped. Rotating the sample gives the full Fermi surface
- Needs UHV ($< 10^{-10}$ torr), clean, flat, well-ordered single-crystalline surface prepared by in-situ cleaving / sputtering-annealing; low temperature for sharp spectra
- Photon energy variation gives k_z dependence (3D bands),

spin-resolved ARPES shows spin texture

- Uses: band structure and effective mass, Fermi surface, superconducting gap (cuprates, iron pnictides), Dirac cone of graphene and topological insulators (Bi_2Se_3), quantum well states in thin films, band bending
- Limitation: very surface sensitive (~ 0.5 – 1 nm), UHV and single crystals, matrix elements and photon energy dependence

II. Diffuse Reflectance Spectra (DRS)

UV-Vis DRS measures light diffusely scattered by a **powder / opaque** sample (transmission is impossible) using an integrating sphere coated with $BaSO_4$ / Spectralon. The reflectance R (λ) vs wavelength is converted into an absorption-like function and used to find the optical band gap.

Kubelka-Munk function

$$F(R) = (1 - R)^2 / 2R = K / S$$

K = absorption coefficient, S = scattering coefficient, R = diffuse reflectance of an infinitely thick sample relative to a white standard.

Tauc plot for the band gap

$$(F(R)\,h\nu)^{1/n} = A\,(h\nu - E_g)$$

- n = 1/2 for direct allowed transition (plot $(F(R)h\nu)^2$ vs $h\nu$); n = 2 for indirect allowed transition (plot $(F(R)h\nu)^{1/2}$ vs $h\nu$). $h\nu = 1240/\lambda(nm)$ eV
- Extrapolate the linear region of the plot to the $h\nu$ axis: intercept = E_g
- Example: TiO_2 anatase $E_g \approx 3.2$ eV (absorption edge ~ 387 nm), ZnO ≈ 3.3 eV; CdS ≈ 2.4 eV
- Blue shift of the absorption edge in smaller particles confirms quantum confinement, red shift / tail absorption shows doping, oxygen vacancies (Urbach energy)
- Uses: photocatalysts, pigments, solar absorbers, dye-sensitised

materials. Limitations: scattering effects, surface reflection, assumptions in K-M theory

12. Photoluminescence (PL) Spectra

PL is the light emitted by a material after it has absorbed photons: **excitation → relaxation → recombination → emission**. It is a contactless and non-destructive probe of the electronic structure, defects and impurities.

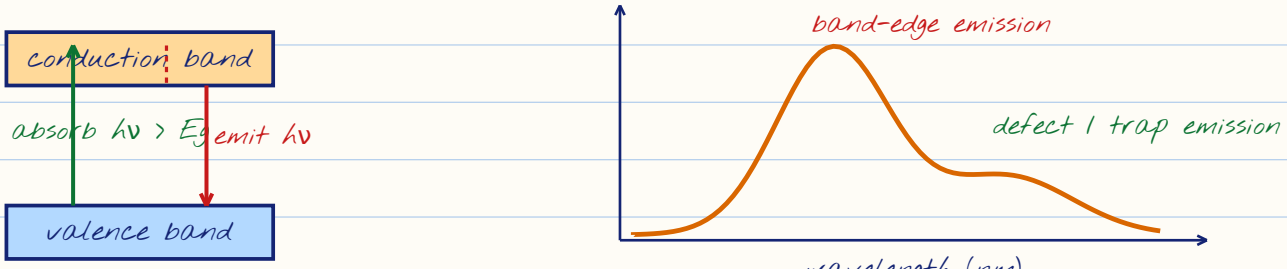


Fig: Principle of PL and a typical emission spectrum

- **Instrument**: Xe lamp / laser (e.g. 325 nm He-Cd) → excitation monochromator → sample → emission monochromator → PMT / CCD detector. Excitation spectrum (fixed emission) and emission spectrum (fixed excitation) are recorded
- **Band-edge (near band edge, NBE) emission** gives E_g; its blue shift shows quantum confinement, FWHM gives size distribution. Stokes shift = absorption − emission energy
- **Defect emission**: oxygen vacancies / zinc interstitials in ZnO give a green-yellow band at 500 – 600 nm; the ratio $I_{UV}/I_{visible}$ measures crystal quality
- **Quenching** of PL (e.g. in TiO_2 / graphene composites) indicates efficient charge separation useful for photocatalysis
- **Time-resolved PL (TRPL)** gives carrier lifetime τ; **quantum yield** = emitted / absorbed photons (> 90 % for CdSe / ZnS dots); temperature-dependent PL gives activation energies and exciton binding energy
- Uses: quantum dots, LED phosphors, solar cell materials (perovskites), biological labels, sensors, defect studies
- Limitation: non-radiative processes are invisible; mostly qualitative

for impurities, interpretation needs care

13. Raman Spectroscopy

Raman spectroscopy is based on <r>inelastic scattering of monochromatic light (laser)</r> by molecular / lattice vibrations (C. V. Raman, 1928, Nobel Prize 1930). The scattered photon gains or loses the energy of a vibration (phonon).

Process	Photon energy change	Intensity
Rayleigh	Elastic, no change	Strongest (blocked by a notch / edge filter)
Stokes Raman	Photon loses energy $\hbar\Omega$ (molecule excited)	Weak ($\sim 10^{-6}$), strong at room temperature
Anti-Stokes Raman	Photon gains energy (from excited vibrational state)	Weaker; rises with temperature, ratio gives T

$$\text{Raman shift } (cm^{-1}) = 1/\lambda_0 - 1/\lambda_s$$

- Selection rule: the vibration must change the **polarisability** of the bond / crystal (Raman active). It complements IR spectroscopy (needs a change in dipole moment); for centro-symmetric molecules mutual exclusion applies
- **Instrument**: laser (488, 514, 532, 633, 785 nm) → microscope → sample → notch filter → spectrometer (grating) → CCD. Micro-Raman spot ~ 1 μm; no sample preparation; works through glass / in water
- Resonance Raman and SERS (surface enhanced, on Au / Ag nanostructures, enhancement 10^6 – 10^{14}, up to single-molecule detection) and TERS give very high sensitivity

Use in nanomaterials

- **Carbon nanotubes**: RBM (radial breathing mode, 100 – 350 cm^{-1}) gives diameter $d(nm) \approx 248/\omega_{RBM}$; G band ~ 1580 cm^{-1} (sp^2); D band ~ 1350 cm^{-1} (defects, I_D/I_G ratio); 2D band ~ 2700 cm^{-1}
- **Graphene**: layer number from the shape of the 2D band (single sharp Lorentzian for monolayer; $I_{2D}/I_G \approx 2$); doping and strain

from G-peak shifts

- **Phonon confinement**: in Si, TiO_2, CdSe nanocrystals the Raman peak shifts and broadens asymmetrically as size decreases; identifies phases (anatase E_g 144 cm^{-1}, rutile 447 / 612 cm^{-1}), stress and defects
- **Others**: MoS_2 layer thickness (E_{2g} – A_{1g} separation), polymorphs of carbon (diamond sharp line at 1332 cm^{-1}), strain in films, composition of alloys
- **Limitations**: fluorescence background, weak signal, laser heating / damage of small particles

Quick Revision – Important Questions

1. Derive Bragg's law. Explain how XRD gives crystallite size by the Scherrer method (with example).
2. Explain the construction and working of SEM; distinguish SEM and TEM.
3. Describe TEM with modes (bright field, HRTEM, SAED) and sample preparation.
4. What is scanning probe microscopy? Explain STM and AFM with diagrams and modes.
5. Explain piezoresponse force microscopy.
6. Write short notes on XPS, XANES / EXAFS and ARPES.
7. Explain DRS and the Tauc plot method for finding the band gap.
8. Explain photoluminescence and Raman spectroscopy with uses in nano-materials.

UNIT - 5

APPLICATIONS OF NANO TECHNOLOGY

Materials - Biology & medicine - Surface science - Energy & environment - Thin films - Quantum dots

1. Overview of Applications

Because of the large surface-to-volume ratio and quantum effects, nanomaterials give better strength, reactivity, conductivity, optical response and selectivity. Their applications cut across every field:

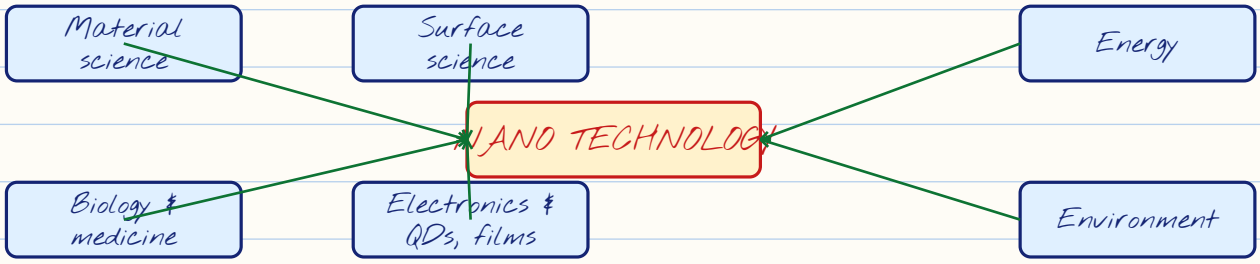


Fig: Fields where nanotechnology is applied

2. Applications in Material Science

2.1 Nanostructured and nanocrystalline materials

- **Nanocrystalline metals and alloys**: grain size 5 - 100 nm gives 2 - 10 times higher hardness / strength (Hall-Petch), better wear and fatigue resistance. Used in cutting tools, gears, bearings, coatings, high-strength steels, Ni-W / Ni-Co electrodeposits
- **Nanoceramics**: nano ZrO_2, Al_2O_3, SiC, Si_3N_4 are tougher, superplastic and sinter at lower temperature (transparent ceramics for armour, bone implants, engine parts, thermal barrier coatings)
- **Hard coatings**: nanocomposite TiN / SiN_x, TiAlN, nc-TiC / a-C with hardness > 40 GPa (superhard), cermets WC-Co with nano WC for drills
- **Magnetic materials**: nanocrystalline Fe-Si-B (Finemet) soft magnets for transformers, Nd-Fe-B nano-composite permanent magnets (exchange-coupled), ferrofluids for seals and speakers

2.2 Nanocomposites

A nanocomposite = matrix + filler where at least one dimension of the filler is < 100 nm.

Type	Examples	Benefits
Polymer matrix (PNC)	Nylon + nanoclay (Toyota timing belt cover), epoxy + CNT / graphene, PET + SiO_2, rubber + carbon black / silica in tyres	Higher strength and modulus at 1 – 5 wt %, barrier to gas (packaging), flame retardancy, conductivity
Ceramic matrix (CMC)	Al_2O_3 + SiC nanoparticles / Al_2O_3 + ZrO_2	Improved fracture toughness, creep and wear resistance
Metal matrix (MMC)	Al + CNT, Cu + graphene, Mg + SiC, Ni + Al_2O_3	High strength to weight, thermal / electrical conductivity

2.3 Smart and functional coatings

- **Self-cleaning** glass (TiO_2: photocatalytic + hydrophilic – decomposes dirt under UV, rain washes it off) and **Lotus effect** superhydrophobic paints (contact angle > 150°)
- **Anti-reflection and anti-fog** coatings (nano porous SiO_2) for lenses and solar panels; **UV blocking** using nano ZnO / TiO_2 in sunscreens, paints, fabrics (transparent as particle < wavelength)
- **Anti-corrosion, anti-scratch** coatings for automobiles, and **anti-bacterial** surfaces with nano Ag, fire retardant nano-silica gels
- **Electrochromic smart windows** (WO_3 films) and thermochromic VO_2 coatings

2.4 Textiles, construction and electronics

- Stain-resistant, wrinkle-free and anti-odour fabrics (nano-whiskers, Ag particles); conductive smart textiles
- Nano-silica and nano-TiO_2 in cement: higher strength, self-cleaning façades, air purifying roads; aerogel insulation
- **Electronics**: shrinking transistor (7 nm / 5 nm node), high-K gate

dielectrics (HfO_2), nano-flash memory, spintronics (GMR, TMR heads), MEMS / NEMS sensors, conductive inks and printed circuits, quantum computing devices

- Nano-lubricants (MoS_2, WS_2, fullerene-like nanoparticles) cut friction; nano-catalysts in refineries; nano-abrasives for polishing (CMP slurry)

3. Applications in Biology and Medicine (Nanomedicine)

Biological units (DNA 2 nm, proteins 5 – 10 nm, viruses 20 – 300 nm, cell organelles) are on the nano scale, so nanoparticles can interact with them directly and can enter cells.

3.1 Drug delivery

- Nanocarriers: **liposomes**, polymeric nanoparticles (PLGA), dendrimers, micelles, gold / silica / magnetic nanoparticles, CNT, nano-gels, albumin-bound (Abraxane)
- **Passive targeting – EPR effect**: tumour blood vessels are leaky (gaps 100 – 800 nm), so 10 – 200 nm particles accumulate in tumours and are retained (poor lymphatic drainage)
- **Active targeting**: surface ligands (antibodies, folic acid, aptamers, peptides RGD) bind to receptors overexpressed on cancer cells. PEGylation ("stealth") prolongs circulation by avoiding immune capture
- **Controlled / triggered release** by pH (acidic tumour 6.5, endosome 5), temperature, enzymes, light, ultrasound or magnetic field
- Benefits: reduces dosage and side effects, protects drug from degradation, crosses the blood-brain barrier, improves solubility of hydrophobic drugs. Examples: Doxil (liposomal doxorubicin), Onpattro (siRNA lipid NP), Pfizer / Moderna COVID-19 mRNA vaccines in lipid nanoparticles

3.2 Diagnosis and imaging

- **Quantum dots** (CdSe/ZnS, InP, carbon dots) as photostable bright fluorescent labels for cells, tissues and in-vivo tumour imaging, multiplexed detection

- **Magnetic nanoparticles** (Fe_3O_4, SPIONs) as MRI T_2 contrast agents (Feridex) and for magnetic separation of cells / DNA; Gd-based nano contrast for T_1 MRI
- **Gold nanoparticles**: lateral flow immunoassay (pregnancy test and COVID antigen strips - red colour from SPR), colorimetric detection of DNA, SERS probes, CT contrast
- **Biosensors and lab-on-a-chip**: nanowire FET sensors, CNT / graphene glucose sensors, nano-pore DNA sequencing (Oxford Nanopore), nanocantilever sensors for proteins, microfluidic point-of-care diagnostics
- **Nano-diagnostics** for early cancer markers (PSA, circulating tumour cells), photoacoustic imaging

3.3 Therapy

- **Photothermal therapy (PTT)**: Au nanorods / nanoshells absorb near-IR (700 - 900 nm, tissue window) and heat tumour to 42 - 50 °C, killing cells. **Photodynamic therapy (PDT)**: nanoparticle photosensitiser generates singlet oxygen
- **Magnetic hyperthermia**: Fe_3O_4 particles heated in an alternating magnetic field (superparamagnetic loss) within tumour (NanoTherm therapy for glioblastoma)
- **Gene therapy**: lipid / polymer nanoparticles carry DNA, mRNA, siRNA, CRISPR into cells; **immunotherapy** nano-vaccines
- **Antimicrobial** nano-silver and copper (wound dressings, catheter coatings, masks), nano-ZnO against resistant bacteria
- **Tissue engineering**: nano-fibre scaffolds (electrospun PCL, collagen) mimic extracellular matrix; nano-hydroxyapatite for bone and dental implants; nano-diamond coatings, titanium nanotube implants; nerve regeneration guides; artificial skin
- **Nanorobots / nano-machines** (concept): DNA origami robots, micro-swimmers for targeted delivery; "theranostics" combine therapy + diagnostics in one particle

Particle	Size	Medical use
Liposome	50 - 200 nm	Drug and vaccine delivery

Gold nanoparticle / nanorod	5 – 100 nm	Diagnostics, PTT, imaging
SPION (Fe_3O_4)	5 – 20 nm	MRI, hyperthermia, magnetic targeting
Quantum dot	2 – 10 nm	Fluorescence imaging, labelling
Dendrimer	2 – 10 nm	Drug and gene carriers
Carbon nanotube / graphene oxide	1 nm dia., 1 µm length	Biosensing, delivery, PTT

Caution:
Nanotoxicity: size, shape, surface charge and coating decide biodistribution, clearance (Kidney clears < 6 nm, larger particles go to liver and spleen) and possible toxicity. Regulation and safety studies are essential before clinical use.

4. Applications in Surface Science

Nanostructuring a surface changes its energy, wettability, reactivity and adhesion. Surface science with nano techniques controls what happens at interfaces.

4.1 Wettability – superhydrophobic and superhydrophilic surfaces

Young: $\cos\theta = (\gamma_{SV} - \gamma_{SL}) / \gamma_{LV}$ Cassie: $\cos\theta^* = f(\cos\theta + 1) - 1$

- Surface roughness at nano / micro scale (lotus leaf: papillae 10 µm covered with wax nano-crystals) traps air (Cassie-Baxter state) so water beads up with $\theta > 150°$ and rolls off with sliding angle < 10° taking dirt away (lotus effect)
- Applications: self-cleaning windows, paints, textiles, anti-icing aircraft wings, anti-fouling ship hulls, oil-water separation meshes, drag reduction, waterproof electronics (liquid repellent nano coating)
- **Superhydrophilic TiO₂** ($\theta \sim 0°$ under UV): anti-fogging mirrors, easily washed
- **Gecko effect**: millions of nano-hairs (setae, spatulae ~ 200 nm)

stick by van der Waals force → dry adhesives, climbing robots

- **Shark skin** riblets reduce drag and bacterial growth. **Moth eye** nano pillars give anti-reflection

4.2 Catalysis and surface reactivity

- Nano-catalysts have a huge number of surface and edge atoms: Au (inert in bulk) becomes highly active for CO oxidation below 5 nm on TiO_2; Pt / Pd nanoparticles on carbon for fuel cells and automobile catalytic converters (reduces use of precious metal)
- Shape-selective catalysts (nano cubes, polyhedra expose particular facets), single atom catalysts, zeolites and mesoporous silica (MCM-41, SBA-15) with 2 – 50 nm pores for petroleum cracking
- Photocatalysts: TiO_2, ZnO, g-C_3N_4 produce electron-hole pairs → reactive radicals (·OH, O_2^-) for water splitting and pollutant degradation

4.3 Self-assembled monolayers (SAM) and surface engineering

- **SAMs**: thiols (R-SH) on Au, silanes (R-$SiCl_3$) on SiO_2. Molecules chemisorb and pack spontaneously in an ordered single layer (1 – 3 nm); end group controls wetting, adhesion, biocompatibility, corrosion resistance
- **Langmuir-Blodgett films** and layer-by-layer assembly build multilayers with nm control
- **Nano-patterning**: dip-pen nanolithography, micro-contact printing, nano-imprint for biochips, sensors, plasmonic substrates
- **Tribology**: nano-lubricants and DLC coatings reduce friction and wear (MEMS, hard disk, engines); **nano-indentation** and nano-scratching test surfaces
- **Sensors**: high surface area increases the adsorption signal (gas sensors from SnO_2, ZnO, WO_3 nanowires detect ppb levels); SERS substrates; QCM
- **Surface plasmon resonance (SPR)** biosensors measure binding of biomolecules by the refractive index change near a thin Au film

Surface effect	Nano-technique	Use

Water repellency	Nano roughness + low energy coating	Self-cleaning glass, fabrics
Adhesion control	Nano hairs, SAMs	Gecko tape, bio-adhesion
Reactivity	High surface area, nano facets	Catalysis, sensing, adsorption
Friction / wear	DLC, MoS_2 nanolayers	Engines, MEMS, hard disk

5. Applications in Energy

5.1 Solar energy

- **Third generation cells**: dye-sensitised solar cells (DSSC) use a mesoporous TiO_2 nano-layer (20 nm particles, 12 μm thick) with Ru dye; efficiency ~ 11 – 13 %.
- **Quantum dot solar cells** (PbS, CdSe, $CuInS_2$): tunable band gap by size, **multiple exciton generation** (MEG: one high energy photon makes 2+ carriers) may exceed the Shockley-Queisser 33 % limit
- **Perovskite solar cells** ($CH_3NH_3PbI_3$ nano-crystalline films, > 26 % efficiency in labs); **thin-film CdTe, CIGS, a-Si**, organic (bulk-heterojunction) cells with nano-phase separated donor / acceptor
- **Light trapping**: Si nanowire / black silicon, plasmonic Au / Ag particles increase absorption in thin films; anti-reflection nano coatings

5.2 Energy storage

- **Li-ion batteries**: nano-$LiFePO_4$, Si nanowire / nanoparticle anodes (accommodate 300 % volume change, capacity 4200 mAh/g vs graphite 372), TiO_2 / $Li_4Ti_5O_{12}$ nano-anodes (fast charge), CNT / graphene conductive additives; shorter Li-ion diffusion path in nano-particles gives high power
- **Supercapacitors**: high surface area activated carbon, graphene, CNT, MnO_2 / RuO_2 nano-oxides (pseudocapacitance) deliver high power density and 10^5 cycles
- **Hydrogen storage**: nano-structured Mg / MgH_2, metal-organic

frameworks (MOFs), CNT, graphene: fast adsorption / desorption kinetics

- **Fuel cells**: Pt nano-catalysts on carbon, nano-structured electrolytes (YSZ thin films) in SOFC, nafion nano-composite membrane

5.3 Energy conversion and saving

- **Thermoelectrics**: nano-structured Bi_2Te_3, PbTe, SiGe, Si nanowires – low lattice thermal conductivity, ZT > 2 for waste heat recovery
- **Piezoelectric nano-generators** (ZnO nanowires) harvest mechanical energy; triboelectric nano-generators for wearables
- **Photocatalytic water splitting** for H_2 (TiO_2, $BiVO_4$, Fe_2O_3); nano-catalysts for CO_2 reduction
- **LED lighting** (GaN nano-structures, QD-LED, QLED TVs) saves energy; **smart windows**, **aerogel** thermal insulation for buildings; lightweight nano-composites for vehicles reduce fuel consumption; efficient nano-catalysts in refineries; high strength wind turbine blades (CNT-epoxy)

6. Applications in Environment

Area	Nano solution
Water purification	Nano-filtration / reverse osmosis membranes, CNT and graphene oxide membranes (desalination), nano-Ag filters for bacteria, nano-zero valent iron (nZVI) for removing Cr^{6+}, As, Pb and chlorinated solvents, magnetic nano-adsorbent (Fe_3O_4) that can be recovered, nano-clays, TiO_2 photocatalysis of dyes and pathogens
Air purification	Nano-TiO_2 coated filters, nano-catalysts in automobile converters (Pt-Pd-Rh) and diesel soot oxidation, nano-fibre HEPA masks, electrostatic filtration, VOC and NO_x removal by photocatalytic paints
Sensing / monitoring	Nano gas sensors for CO, NO_2, H_2S, VOCs; heavy metal sensors with Au nanoparticles (colorimetric); biosensors for pathogens in water; SERS pesticide detection

Remediation of soil	nZVI injection in groundwater plumes; nano-bioremediation; nano-fertiliser and nano-pesticide for precision agriculture (reduce chemical use)
Green manufacturing / waste	Nano-catalysts lower energy and waste; recyclable nano-adsorbents; oil spill clean-up using magnetic / hydrophobic sponges (graphene sponge, CNT aerogel)
Climate	CO_2 capture by MOFs / zeolites / nano-amines; photocatalytic CO_2 conversion; lightweight materials to cut emissions

Caution:
Environmental risk: engineered nanoparticles may accumulate and be toxic to organisms; life cycle assessment and safe-by-design are required.

7. Applications of Nanostructured Thin Films

Nanostructured thin films have thickness or grain size / porosity in the nanometre range (single layer, multilayer, superlattice, nano-composite, nano-porous), prepared by PVD / CVD / ALD / sol-gel / spin-coating, etc.

Application	Film / structure	Function
Optical coatings	Multilayer SiO_2 / TiO_2 (quarter-wave stacks), nano-porous SiO_2	Anti-reflection, dielectric mirrors, filters, beam splitters, low-E and heat-reflecting glass, Bragg reflectors in VCSEL
Solar cells	a-Si:H, CdTe, CIGS, perovskite, TCO (ITO, AZO, FTO)	Absorber and transparent electrodes, thin and flexible
Microelectronics	High-K HfO_2 (ALD), Si_3N_4, Cu barrier TaN, ferroelectric PZT films (FeRAM)	Gate dielectric, interconnects, memory, capacitors
Magnetic storage	Co/Pt, Fe/Cr multilayers, spin valves, MTJ (MgO), CoCrPt-SiO_2 granular films	GMR / TMR read heads, hard disks (> 1 Tb/in²), MRAM

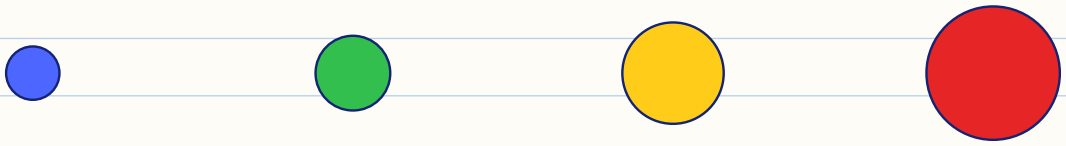
Sensors	SnO_2, ZnO, WO_3 nano-porous films; SAW devices (AlN, ZnO piezo films)	Gas, humidity, bio-sensing, MEMS
Hard / protective	TiN, TiAlN, CrN, DLC, nano-multilayers (TiN/AlN)	Cutting tools, wear and corrosion protection, decorative
Display and lighting	OLED organic multilayers, ITO, QLED, thin-film transistors (a-Si, IGZO)	Flat panel displays, flexible screens, solid state lighting
Energy storage	Thin-film Li batteries (LiPON electrolyte), electrochromic WO_3, supercapacitor films	Micro-batteries for IoT, smart windows
Superconducting	YBCO, Nb films, Josephson junctions	SQUID, qubits, RF filters
Thermoelectric	Bi_2Te_3 / Sb_2Te_3 superlattices	Micro-coolers, energy harvesting
Biomedical	Hydroxyapatite on implants, antibacterial Ag-TiO_2, DLC on stents	Biocompatibility, bone integration

- **Quantum wells and superlattices** (GaAs/AlGaAs) are the active layers in laser diodes, HEMT transistors, quantum cascade lasers, infrared photo-detectors
- **Multilayer films** utilise interference (optical), spin-dependent scattering (GMR) or phonon blocking (thermoelectric) effects only possible when layer thickness is in nm
- Dependence: properties depend on thickness, strain, interface, substrate and grain size; thin-film deposition control (ALD: 0.1 nm per cycle) is the Key

8. Applications of Quantum Dots

A quantum dot (QD) is a semiconductor nanocrystal (2 - 10 nm) in which carriers are confined in all three dimensions, so it has discrete atom-like energy levels and size-tunable band gap.

smaller dot → larger band gap → blue emission



CdSe 2 nm
blue ~ 480 nm

CdSe 3 nm
green ~ 530

CdSe 4.5 nm
yellow ~ 580

CdSe 6 nm
red ~ 630 nm

Fig: Size dependent colour of CdSe quantum dots (emission under UV)

Properties that make QDs useful

- Size-tunable absorption / emission, narrow emission (FWHM 20 - 30 nm), broad absorption, high quantum yield (> 90 %), photostability (100 - 1000 times more than organic dyes), long lifetime (10 - 30 ns), solution processable and low-cost
- Core-shell structure (CdSe/ZnS) reduces surface traps; non-toxic QDs: InP, $CuInS_2$, ZnSe, Si, graphene / carbon dots, perovskite ($CsPbBr_3$)

Field	Application of quantum dots
Displays	QLED / quantum-dot enhanced TVs (QD film on blue LED gives pure R, G, B, wider colour gamut > 100 % NTSC), electroluminescent QD-LED, micro-LED colour conversion
Lighting	White LEDs with QD phosphor: warm light, high CRI, lower energy; horticulture lighting
Solar cells	QD sensitised and QD solar cells (PbS, PbSe), luminescent solar concentrators (QD absorbs and re-emits to edge cells), MEG, tandem cells, intermediate band cells
Lasers and photonics	QD laser diodes (InAs/GaAs): low threshold current, temperature insensitive, used in optical communication; single photon source for quantum cryptography; optical amplifier
Photodetectors / imaging	PbS QD infrared detectors and cameras (SWIR), CMOS image sensors, night vision, thermal imaging
Bio-medical imaging	Fluorescent labels, in-vivo tumour imaging, FRET sensors, flow cytometry, multiplexed immunoassay, photodynamic therapy, drug tracking; image-guided surgery

Sensors	Fluorescence quenching sensors for pH, heavy metal, toxins, explosives; QD gas sensors
Electronics and computing	Single electron transistor (SET), quantum dot memory, QD cellular automata, spin / charge qubits in quantum computers (Loss-DiVincenzo)
Security / anti-counterfeiting	Invisible QD inks and tags (currency, documents), barcoding
Catalysis / environment	QD photocatalytic H_2 evolution, CO_2 reduction, dye degradation

- **Limitations**: toxicity of Cd / Pb based dots, blinking (intermittent fluorescence), aggregation and stability in water, cost of large-scale monodispersed production, regulation (RoHS)
- 2023 Nobel Prize in Chemistry (Bawendi, Brus, Ekimov) honours discovery and synthesis of quantum dots

Quick Revision - Important Questions

1. Discuss the applications of nanotechnology in material science (nanocomposites, coatings).
2. Explain nanomedicine: drug delivery, diagnostics and therapy with examples.
3. Explain the applications of nano-technology in surface science (lotus effect, catalysis, SAM).
4. How is nanotechnology useful in energy generation and storage? Discuss solar cells, batteries, supercapacitors.
5. Describe applications of nanomaterials in environmental protection (water and air purification).
6. Write about the applications of nanostructured thin films.
7. What are quantum dots? Write their properties and applications.